

Bernoulli trial.

Many random experiments that we carry have only two outcomes that are either **failure** or **success**. For example, a product can be defective or non-defective, etc. These types of independent trials which have only two possible outcomes are known as Bernoulli trials.

For the trials to be categorized as Bernoulli trials it must satisfy these conditions:

1. A number of trials should be finite.
2. The trials must be independent.
3. Each trial should have exactly two outcomes: success or failure.
4. The probability of success, denoted by **P** remains constant from trial to trial.

Example of Bernoulli Trials

Eight balls are drawn from a bag containing 10 white and 10 black balls. Predict whether the trials are Bernoulli trials if the ball drawn is replaced and not replaced.

(a) For the first case, when a ball is drawn with replacement, the probability of success (say, white ball) is $p=10/20=1/2$ which is same for all eight trials (draws). Hence, the trial involving drawing of balls with replacements are said to be Bernoulli trials.

b) For the second case, when a ball is drawn without replacement, the probability of success (say, white ball) varies with the number of trials. For example for the first trial, probability of success, $p=10/20$ for second trial, probability of success, $p=9/19$ which is not equal the first trial. Hence, the trials involving drawing of balls without replacements are not Bernoulli trials

Binomial Distribution

Consider Bernoulli trials for tossing a coin 10 Times. Let obtaining head, stand for success, **P** and tails for failure, **T**.

probability of success, $p = \frac{1}{2}$ (Constant)

probability of failure , $q = \frac{1}{2}$ (Constant)

Where $p + q = 1$

We have following situations tossing coins 10 times

$p p q q p p q q p p$ **$n = 10$**

6 times success, (10-6) times failure, possible combinations for success are **${}^{10}C_6$**

$${}^{10}C_6 P^6 q^{10-6}, x = 0, 1, 2, \dots, 10$$

For n trials, number of ways for x successes, T and $(n - x)$ failures, q can be given as

$p p q q p p q q p p \dots \dots \dots p q p$ (n times)

$$n C_x P^x q^{n-x}, x = 0, 1, 2 \dots \dots \dots n$$

A Bernoulli trial can result in a success with probability p and a failure with probability $q = 1-p$. Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

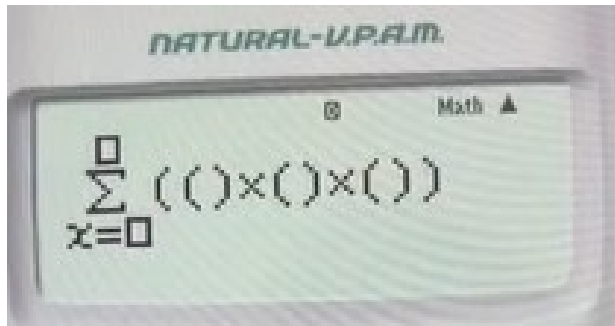
$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

Calculator use -991ES

$$B(x, n, p) = \sum_{x=0}^{10} \binom{n}{x} p^x q^{n-x}$$

For example

$$p = 0.30, \quad q = 1 - 0.30, \quad n = 10$$



$b(x; n, P)$: Binomial probability - the probability that an **n-trial** binomial experiment results in exactly **x successes**, when the probability of success on an individual trial is P .

Example 5.1:

The probability that a certain kind of component will survive a shock test is $3/4$. Find the probability that exactly 2 of the next 4 components tested survive.

Assuming that the tests are independent and $p = 3/4$ for each of the 4 tests, we obtain

$$b\left(2; 4, \frac{3}{4}\right) = \binom{4}{2} \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 = \left(\frac{4!}{2! 2!}\right) \left(\frac{3^2}{4^4}\right) = \frac{27}{128}.$$

When working the binomial distribution we will often need to find the probability of there being at least r success ,

$$P(X \geq r) = 1 - P(x < r)$$

Example

A student is taking a multiple choice quiz but forgot to study and so he will randomly guess the answer to each question. There are a total of 12 questions, each with 4 answer choices. Only one answer is correct for each question

Solution

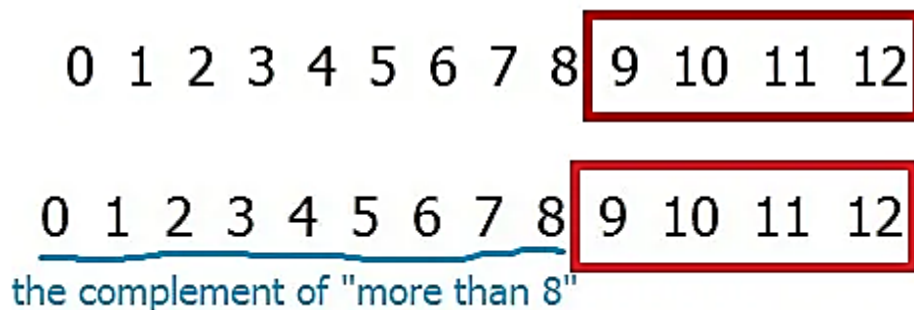
we have $n=12$ ". There are two outcomes: **"guess correctly", "guess incorrectly"**.

If we treat a success as guessing a question correctly, then since there are 4 answer choices and only 1 is correct, the probability of success is:

$$P = \frac{1}{4} = 0.25$$

(a) Find the probability that he correctly answers more than 8 questions.

This probability is represented by $P(X > 8)$. To understand how to find this probability using `binomcdf`, it is helpful to look at the following diagram.



$$P(X > 8) = 1 - P(X < 8)$$

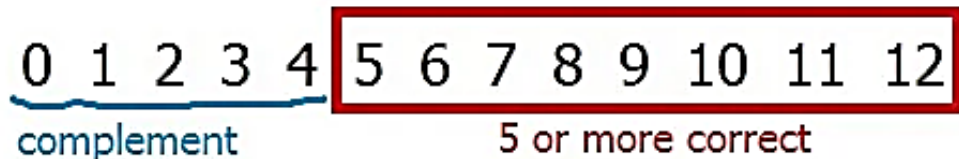
$$1 - \sum_{k=0}^8 \binom{12}{k} (0.25)^k (1 - 0.25)^{12-k} = \mathbf{0.00278}$$

Method 2

$$= \sum_{x=8}^{12} \left[(12Cx) \times (0.25)^x \times (0.75)^{12-x} \right]$$
$$= \mathbf{0.00278}$$

(b) Find the probability that he correctly answers 5 or more questions.

This probability is represented by $P(X \geq 5)$. Using our diagram:



$$P(x \geq 5) = 1 - P(x < 5)$$

$$= 1 - \sum_{x=0}^4 \left[({}^{12}C_x) \times (0.25)^x \times (0.75)^{12-x} \right]$$

$$= 0.1576$$

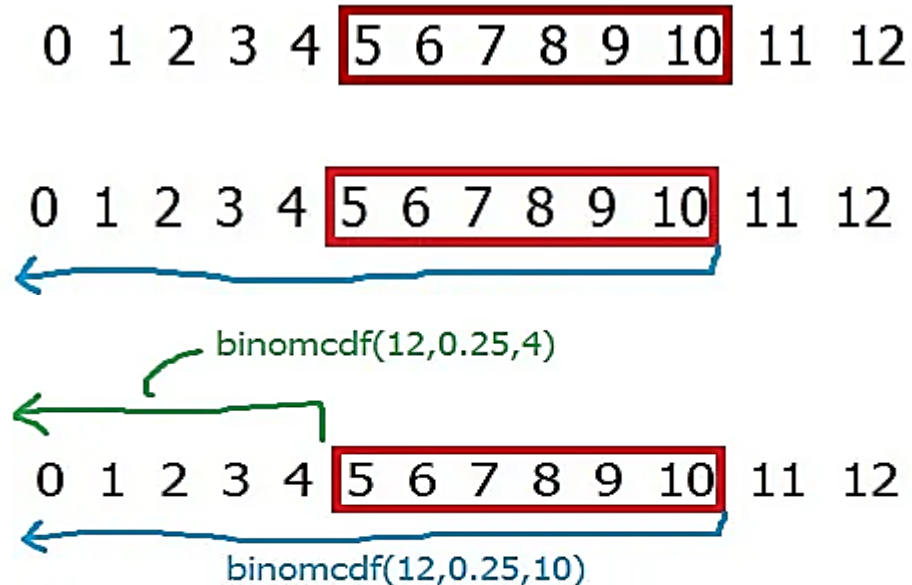
Method 2

$$= \sum_{x=5}^{12} \left[({}^{12}C_x) \times (0.25)^x \times (0.75)^{12-x} \right]$$

$$= 0.1576$$

(c) Find the probability that he correctly answers between 5 and 10 questions (inclusive) correctly.

Since this is inclusive, we are including the values of 5 and 10. That is, we are finding $P(5 \leq X \leq 10)$.



$$= \sum_{x=0}^{10} \left[(12Cx) \times (0.25)^x \times (0.75)^{12-x} \right] - \sum_{x=0}^4 \left[(12Cx) \times (0.25)^x \times (0.75)^{12-x} \right]$$

$$= 0.1576$$

Method 2

$$= \sum_{x=5}^{10} \left[(12Cx) \times (0.25)^x \times (0.75)^{12-x} \right]$$

$$= 0.1576$$

Example 5.2:

The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that

- (a) at least 10 survive
- (b) from 3 to 8 survive.
- (c) exactly 5 survive?

Solution

Probability Of Success = $P = 0.4$

Failure = $q = 1 - 0.4 = 0.6$

$n = 15$

Solution: Let X be the number of people who survive.

$$\begin{aligned} \text{(a)} \quad P(X \geq 10) &= 1 - P(X < 10) = 1 - \sum_{x=0}^9 b(x; 15, 0.4) = 1 - 0.9662 \\ &= 0.0338 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(3 \leq X \leq 8) &= \sum_{x=3}^8 b(x; 15, 0.4) = \sum_{x=0}^8 b(x; 15, 0.4) - \sum_{x=0}^2 b(x; 15, 0.4) \\ &= 0.9050 - 0.0271 = 0.8779 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad P(X = 5) &= b(5; 15, 0.4) = \sum_{x=0}^5 b(x; 15, 0.4) - \sum_{x=0}^4 b(x; 15, 0.4) \\ &= 0.4032 - 0.2173 = 0.1859 \end{aligned}$$

Example 5.3:

A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

(a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?

Solution

Denote by X the number of defective devices among the 20. Then X follows $b(x; 20, 0.03)$ distribution. Hence,

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) = 1 - b(0; 20, 0.03) \\ &= 1 - (0.03)^0 (1 - 0.03)^{20-0} \\ &= 0.4562 \end{aligned}$$

Example

Each sample of water has a 10% chance of containing a particular organic pollutant. Assume that the samples are independent with regard to the presence of the pollutant. Find the probability that in the next 18 samples, exactly 2 contain the pollutant

Solution

Let X the number of samples that contain the pollutant in the next 18 samples analyzed.

Then X is a binomial random variable with $p = 0.1$ and $n = 18$.

Therefor

$$P(X = 2) = \binom{18}{2} (0.1)^2 (0.9)^{16}$$

(b) Determine the probability that at least four samples contain the pollutant.

$$\begin{aligned}P(X \geq 4) &= 1 - P(X < 4) = 1 - \sum_{x=0}^3 \binom{18}{x} (0.1)^x (0.9)^{18-x} \\&= 1 - [0.150 + 0.300 + 0.284 + 0.168] = 0.098\end{aligned}$$

(c) Determine the probability that $3 \leq X < 7$

$$\begin{aligned}P(3 \leq X < 7) &= \sum_{x=3}^6 \binom{18}{x} (0.1)^x (0.9)^{18-x} \\&= 0.168 + 0.070 + 0.022 + 0.005 \\&= 0.265\end{aligned}$$

If X is a binomial random variable with parameters p and n ,

$$\text{mean} = E(X) = np, \quad \text{variance} = V(X) = np(1 - p)$$

Find the mean and variance of the binomial random variable of Example 5.2,

$$\mu = (15)(0.4) = 6 \text{ and}$$

$$\sigma^2 = (15)(0.4)(0.6) = 3.6.$$

QUESTION

An electronic product contains 40 integrated circuits. The probability that any integrated circuit is defective is 0.01, and the integrated circuits are independent. The product operates only if there are no defective integrated circuits. What is the probability that the product operates?

$$n = 40,$$

The probability that any integrated circuit is defective = 0.01 here $P = 0.01$